STAFF REVIEW · H446 258 source models

Separate from the eight-page pupil workbook. This copy is in a public repository and is NOT access controlled. Check technical correctness before use.

Printed question mapping: Booklet Q1 = original online Q1; Booklet Q2 = original online Q2; Booklet Q3 = original online Q3

Do Now — source answers

1. Converting an analogue sensor signal into a digital representation.

2. CIR.

3. The next step cannot start until the earlier result exists.

4. $1+1+0=10_2$; sum bit 0, carry-out bit 1.

I Do — source worked steps

1. Predict — Copy PC=40 into MAR to select the instruction location.

2. Explain — Instruction contents travel via MDR to CIR, while sequential PC advances to 41.

We Do — guided CHECK

1. Predict
Each available core receives a separate independent job.
Explain
Ideal duration is the longer job, five minutes; serial execution would take eight.

You Do — source mark schemes

1. [5] PC next instruction address [1], MAR memory address [1], MDR transferred data/instruction [1], CIR current instruction [1], ACC interim arithmetic result [1].

2. [6] Cache can reduce some slow main-memory accesses [2]; cores can process independent frames or tiles [2]; dependencies, bandwidth and overhead reduce ideal gains [2].

3. [6] General control with branching suits CPU [2]; similar pixel calculations suit many GPU lanes [2]; data transfer and parallelisation overhead determine actual suitability [2].

Exit Ticket — source answer

1. False. Only suitably parallel work can benefit and overhead limits scaling.